

StraboField Tephra User Guide

Developed spring 2026 for first public release of StraboField Tephra module

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Introduction

StraboField is a flexible mobile app designed to facilitate field data collection for a diverse array of geoscience applications. Several modules exist within StraboField for structural data, tectonic stratigraphy, mineral alteration, and other disciplines. Here we present a new module developed specifically for tephra stratigraphy. The module was designed following guidelines developed in the tephra community best practices working group (Wallace et al. 2022) and includes the standardized terminology and field documentation recommended by this effort. While optimized for work related to volcanic tephra studies, the structure and workflow may be useful for many Quaternary soft-sediment stratigraphic applications. The module will be especially useful to students new to the field. Importantly, documenting stratigraphic data with this app provides backend digitized data that are compatible with data publication and database development.

The core concept of the StraboSpot system (which StraboField is a part of) is the concept of nested “spots”. For tephra stratigraphic work, we have adapted this system such that:

- Each stratigraphic section is a “**station**”, which is recorded as a spot on the basemap with a latitude-longitude, location description, and unique name (hereafter called an ID).
- Each described stratigraphic “**layer**” in a section is a spot nested underneath the station spot. By convention, these layers inherit the station ID as a prefix and each layer is numbered in sequence from top to bottom (or bottom to top). Single tephra deposits can be broken into separate layers to describe complex sequences, such as mixed normal and reverse grading. Similarly, it is often useful to describe a well-preserved tephra as a separate layer than the soil developed on top of the tephra, rather than lumping both

together as a single “tephra-soil complex”. Alternatively, at times it is necessary to lump complex sequences as a single “undifferentiated” package of deposits where individual units are not described due to time constraints in the field.

- The Sample module of StraboField is useful for documenting physical “**samples**” collected from a tephra layer (or just from the station itself in the case of a lava, ballistic, or edifice samples).

Within StraboField, there are two fundamental ways to document a tephra section: (1) annotating layers on a photograph of the section, (2) listing layers without an image. We feel (1) is the most powerful way to use the app, clearly indicating where layers you describe are with a real image of the section, but will also describe option (2) which can be handy in some situations.

Getting Started

Please refer to the StraboField user guide for comprehensive documentation (<https://strabospot.org/manual/field>). For quick start information for the tephra module:

Registering a StraboSpot account

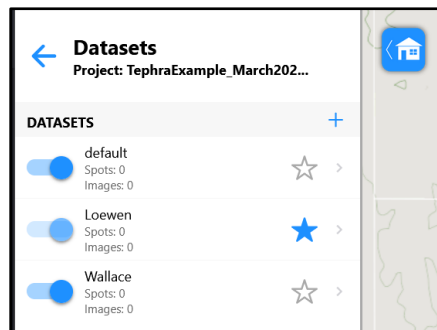
On your computer in a web browser:

Register a username and password at <https://strabospot.org/register>; accounts are free! If working on a collaborative project, it is useful to setup a shared account and all use the same username/password but store data in different datasets (more later).

Setting up a new project

With an account setup, it is now time to set up a new project. This is accomplished easiest using the online interface and a computer.

1. Click on your username (top right of page) and select “My StraboField Data”
2. “Add Project” and fill out the form
3. After creating the project, click on the “Options” button underneath its new header and choose– View/Edit/Add data – This will bring you to a web-based version of the mobile app.
4. The Home button in the top left brings up menu options:
 - Under “Datasets”, add datasets for each field team member collecting separate data.



5. Naming conventions has some useful features for tephra site descriptions, we recommend the following setup, but users can modify to their preference:
- Use a consistent prefix convention, the Alaska Volcano Observatory (AVO) standard is Year-Volcano Initial-Geologist Initial, e.g., 26VOML01, 26VOML02, 26VOKW01, 26VOKW02. Placing “26VOML” in “prefix” and “1” in “Starting number”
 - You will likely name layers on an image basemap, so it is helpful to have these “prepend” the parent spot name and then number each layer on an image starting with 1. Also add “-” in the alternate prefix field.
 - If you record samples with the app, you will want to prepend the spot name (first toggle) and add a letter for multiple samples (vs. number for layers) (last toggle).
 - We suggest enabling warning to prevent accidentally making duplicate spot or sample names.

Naming Conventions
Project: TephraExample_March202...

SPOT NAME

Prefix ⓘ
26VOML

Starting Number ⓘ
1

☒ Warn on Duplicate Spot Name ⓘ

IMAGE BASEMAP / STRAT SECTION SPOT NAME

Alternate Prefix ⓘ
-

☒ Prepend Parent Spot Name ⓘ

☒ Restart at 1 for each Parent Spot

SAMPLE NAME

Prefix ⓘ
Short label based on location or IGSN prefix (exactl

Starting Number ⓘ
Number to append to the end of every sample nan

☒ Prepend Spot Name ⓘ

☒ Warn on Duplicate Sample Name ⓘ

☐ Restart at 1 for each Spot

☒ Append Letter Instead of Number to End of Name, Restarting at 'a' for each Spot ⓘ

6. Uploading custom basemaps (e.g., geologic maps, non-default imagery or elevation products).
- Back on the main StraboSpot page (previous browser tab) you will need to upload the basemaps to Strabo to become available in the project

- Click on your username (upper right) and select “My Maps”
 - Choose “+Add new map”
 - Select a geotiff file (ideally compressed to a reasonable file size of <<1 GB and in WGS84 coordinates)
 - This uploads and displays a map code (“My Maps ID”) that you will put into your project to access the map.
7. Adding custom basemaps to your project:
- Back in the webapp, under the home menu, scroll down to the Maps section and choose “Custom Maps” > “+ Add New Custom Map”
 - Give the map a useful name and choose “Strabospot My Maps” and paste in the My Maps ID
 - Do NOT select “Display as overlay”

← Custom Maps

Add Map

CUSTOM MAP TITLE

CustomDEM

OVERLAY SETTINGS

To save this map as an overlay for offline use first save as a basemap then switch it to an overlay.

Display as overlay ☐

MAP TYPE

If you wish to save a new MapWarper map please download the .tiff file from Mapwarper.net and upload it into your Strabo MyMaps account.

Mapbox Styles ☐

Strabospot My Maps ☒

MAP DETAILS

Strabo My Maps ID

Map ID is required

Using the tephra module

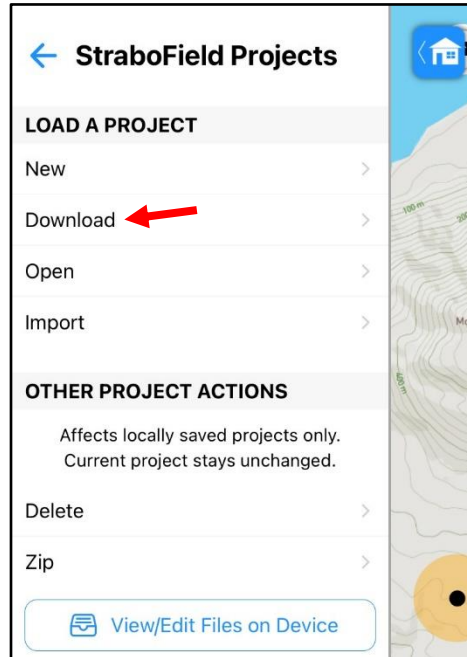
Installing the StraboField application

On your mobile device:

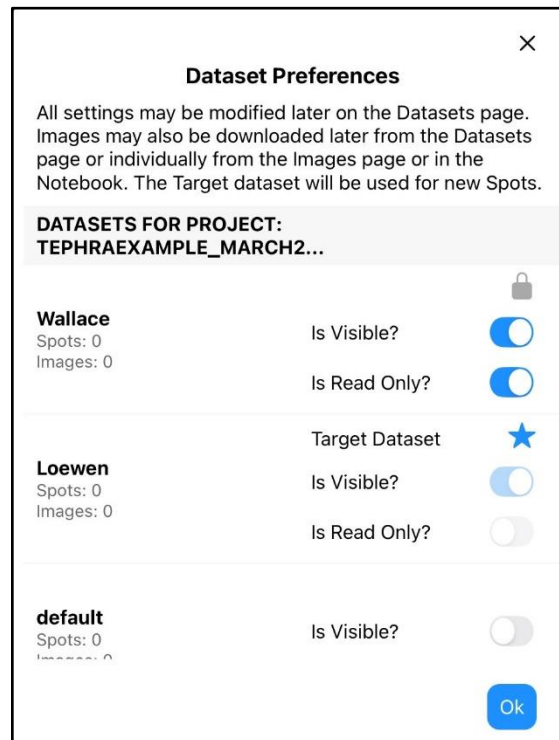
1. Download StraboField to your mobile device from the Apple App Store or Google Play Store.
2. Now that your project is setup online, on your mobile device open the StraboField app and login to the same account.

Downloading a project from the server

1. Using the Home menu, scroll down to the ACCOUNT heading and select “StraboField Projects” > “Download” and download the project you setup online (note, you can setup a project inside the mobile app, too).

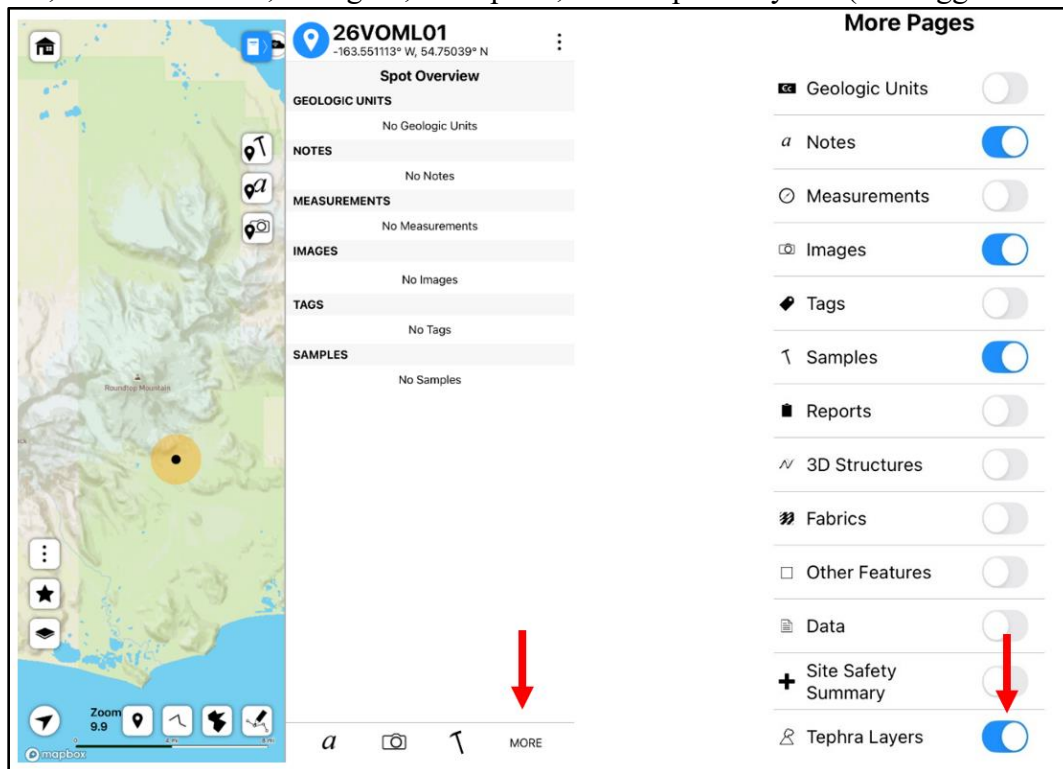


2. Make any desired datasets visible and set yours to “Target” (star). Suggest making other datasets “read only”.



Verifying your project works

1. Make your first “station” by placing a spot on the map with the pin symbol tool in the lower left (upside down teardrop). You can also long-press the spot tool (or Home>App Settings>Adding New Spots>”Add to Current Location”) to toggle placing the spot automatically at your current location.
2. This will open your notebook (right side menu) with this first spot. At the bottom right go to “MORE” to select the notebook pages we want to have. For the tephra module as we describe, turn on “Notes”, “Images”, “Samples”, and “Tephra Layers” (and toggle others off).



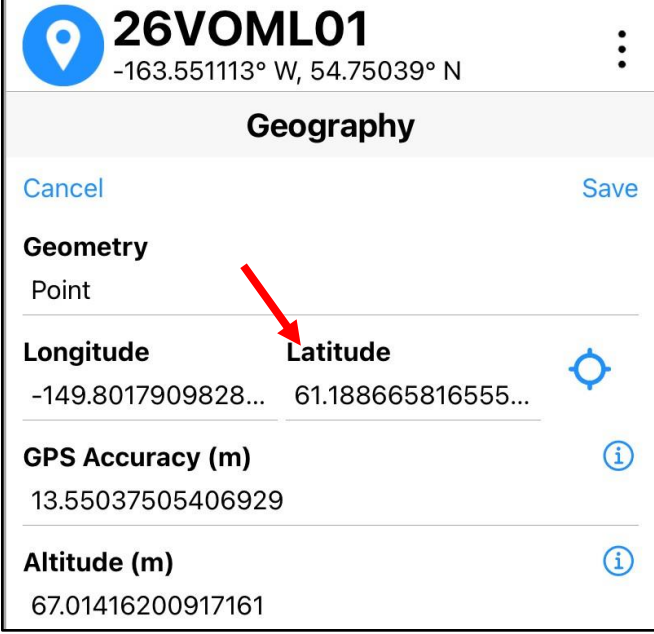
3. If this was just a practice spot, go ahead and delete this spot (vertical three dots, top right), and in the left Home menu, scroll to “Customize and Preset” > “Naming Conventions” and reset spot numbers to start at 1 again.
4. Before you go in the field, you will likely want to save basemaps for offline use. See [StraboField manual](#) for detailed information, but the quick version is:
 - In the app on your mobile device, tap the layers icon in the lower left on the map and choose a basemap that you want to be available offline (one of the defaults or a custom one). This will display in the map view.
 - Zoom the map extent to the smallest extent that you want to save (e.g., choose the minimum area to download to avoid downloading data outside of your area of interest). If you choose a large area, the app will restrict how high of a zoom level (aka resolution) that you can download.
 - Tap on the three vertical dots in the lower left to open the menu of “Map Actions” and choose “Save Map for Offline Use”

- Choose the desired “zoom” level and select “Download”.
 - For large areas you may have to iterate your map view area, panning to subsections of the map at a high enough zoom level to download the needed resolution.
 - Turn off wi-fi and turn airplane mode on, to test functionality before going in the field!
5. You are now ready!

Collecting Station-Layers-Sample data

Stations

1. As in the test spot, use the spot tool to place a “station” on the map, or hold down the spot tool to set the station to your current location (pin icon with compass arrow). If you place a station on the map and want to have it change to your current location, simply tap on the lat-long coordinates below the station name and use the target icon to set it to your current location, this will also show you your GPS accuracy (try again if accuracy is poor, sometimes GPS takes a while to improve its location).



 **26VOML01** 
-163.551113° W, 54.75039° N

Geography

[Cancel](#) [Save](#)

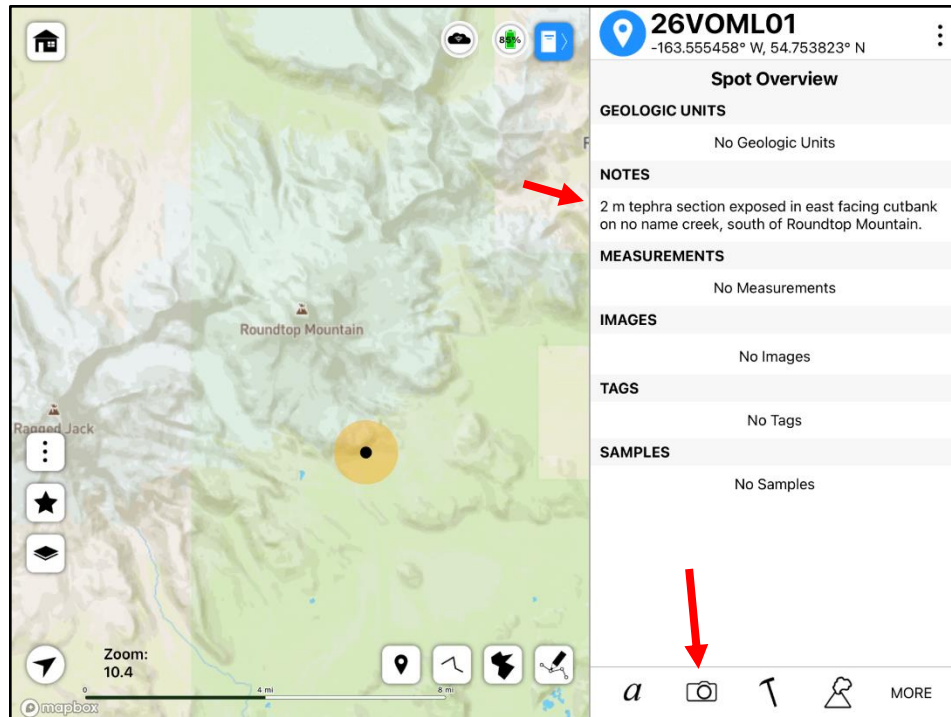
Geometry
Point

Longitude	Latitude	
-149.8017909828...	61.188665816555...	

GPS Accuracy (m) 
13.55037505406929

Altitude (m) 
67.01416200917161

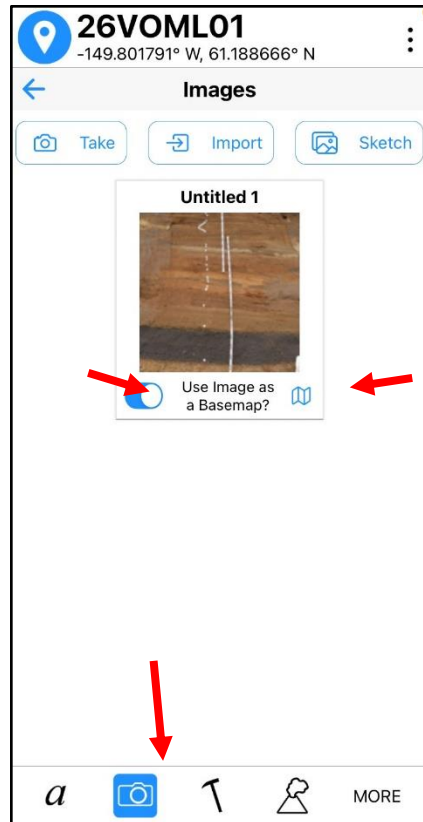
2. You can use the “a” icon to access the Notes tab and add a description of the location, and any other observations you want to make. We also use stations without samples or stratigraphic sections to describe general observations during a field project.
3. You can use the camera icon to go to the Images tab, and take photos directly from the app, important photos from your camera roll, or make a sketch (Apple pencil or similar recommended). These are saved within the spot to add visual context to the location, samples, or stratigraphy.



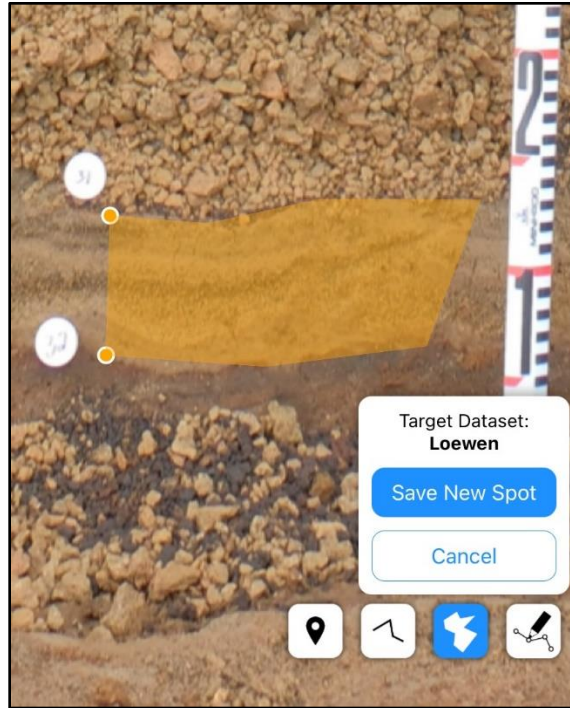
4. For edifice sampling, we can add a sample directly to the station without entering the tephra module by selecting the hammer icon to go to the “Samples” tab

Tephra Layers Option 1 (annotate an image)

1. Add photo(s) or even a sketch of the tephra section using the camera icon. You can take a picture with our device or import a picture you already took (or took with a camera and transferred to your device). You can even sketch the section and add that.
2. If you tap on the photo it will open in an image viewer, with the ability to add metadata to the image (scale information, viewing geometry, caption, etc). If you tap on the photo icon with the pencil, you can also sketch on the image itself (which is then saved as a new image in the spot)
3. If you want to log your tephra layers directly on an image (highly recommended), toggle on “Use Image as a Basemap” below the thumbnail.



4. Click the map icon to the right of the basemap toggle to open the image.
5. Now the geographic map view in the main window is replaced by the image. You can therefore now add spots into the image (nested under the parent “station” spot) based on layers in the image.
6. Use the polygon tool (lower right) to outline the extent of a layer (suggest working systematically top to bottom or bottom to top). Save as new spot. Note you can zoom in on the image before starting to annotate the layer. You can also use a line or point, but for layers the polygon can most clearly capture the zone of the imager you are describing.



7. This will open a new spot that now has a default name after the preset naming convention
8. In the notebook window (right side), use the volcano icon to switch to the Tephra layers tab, and + to add layer metadata. This opens up the “Basic” tab with the following fields:
 - Layer Label gives the layer a name. This is most important for layers described without an image (say, using numbered layers). Here, the spot is your label.
 - Select Layer Type from list (REQUIRED).

26VOML01-01
Polygon

Tephra Layer Detail

Basic Additional

Cancel Save

Layer Label
26VOML01-01-1

Layer Type
tephra fall

- tephra fall ✓
- pyroclastic density current
- pyroclastic surge
- pyroclastic flow
- sediment
- soil
- peat
- lacustrine
- fluvial
- eolian
- diamict
- mass wasting
- debris flow
- lahar

a MORE

- Describe layer color (optional text field).
- Layer thickness fields
 - Select the units for describing thickness, should default to centimeters
 - Describe 1 or more of the numeric thickness fields: The “typical” thickness, a “minimum” thickness, and/or a “maximum” thickness. ONE of these fields is REQUIRED, more are optional.
 - a) If the layer is uniform, just describe “typical”
 - b) If the layer is variable, describe “minimum” and “maximum”. Note a discontinuous layer could have “0” for the minimum.
 - c) If the layer is variable, but had a most common thickness you want to note that is not the average, also include “typical”
 - d) If the unit is not completely exposed (bottom of section), you might want to only include the minimum thickness and leave other fields blank.
- Grain size fields
 - Select a grainsize to describe the bottom of the layer (optional but strongly encouraged). We suggest describing the “maximum mode” for a poorly sorted deposit
 - Select a grainsize to describe the top of the layer (optional but strongly encouraged).

- Select a choice to describe the grading (optional, note the choice of bottom/top grainsize might already define as normal or reverse, be sure your selection is consistent).
- Select a choice to describe the grainsize sorting (optional).

26VOML01-01
Polygon

Tephra Layer Detail

Basic Additional

Cancel Save

Layer Label
26VOML01-01-1

Layer Type
tephra fall

Layer Color
Tan

Thickness Units
cm

Typical Thickness
7

Minimum Thickness
6

Maximum Thickness
10

Bottom Grainsize
fine ash/sand (1/8-1/4 mm)

Top Grainsize
very coarse ash/sand (1-2 mm)

Grading
Reverse

Sorting
well sorted

Delete Tephra Layer

a MORE

- Hitting save now will exit out of the layer detail panel!
- Go to the “Additional” tab (top toggle)
 - Select values for the top and bottom contacts (optional).
 - Select a value for the grain support (optional).
 - Select one or more values for grain angularity (optional).
 - Select a choice for tephra concentration (optional).
 - Use the “Layer Description” to describe anything not in one of the defined fields, especially useful for describing non-standard items like componentry, clast size measurements, or mineralogy (optional).
 - List the names of any volcanoes you think might be the source of a tephra (optional, and only applicable to tephra layers).
 - Give the name of the tephra, either a formal geologic name (“1912 Katmai”) or an informal field name used to correlate between sections (“Tephra A”).

26VOML01-01
Polygon

Tephra Layer Detail

Basic Additional

Cancel Save

Top Contact
sharp

Grain Support
clast

Grain Angularity
Multiple Selected (2 selected)
subangular angular

Tephra Concentration
discrete

Layer Description
Variable layered fine deposit found at the Fisher base.

Possible Source Volcano(es)
Fisher

Tephra Name
Fisher CFE

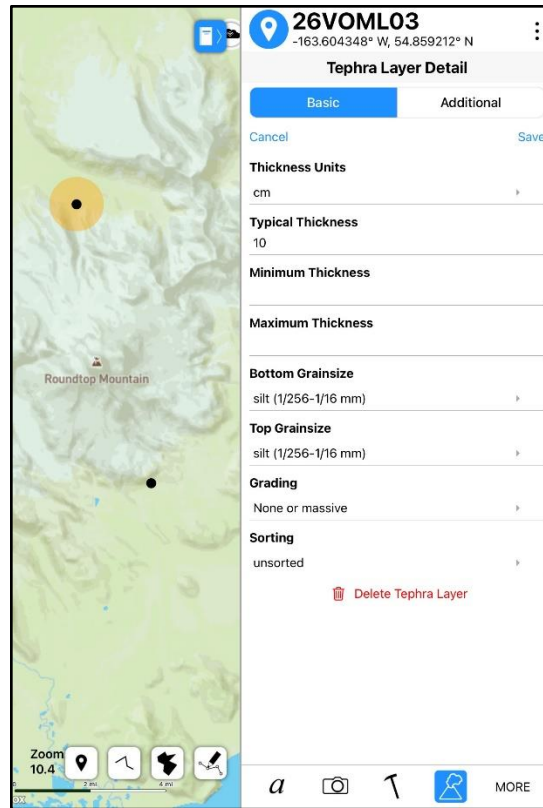
Delete Tephra Layer

α camera pencil person MORE

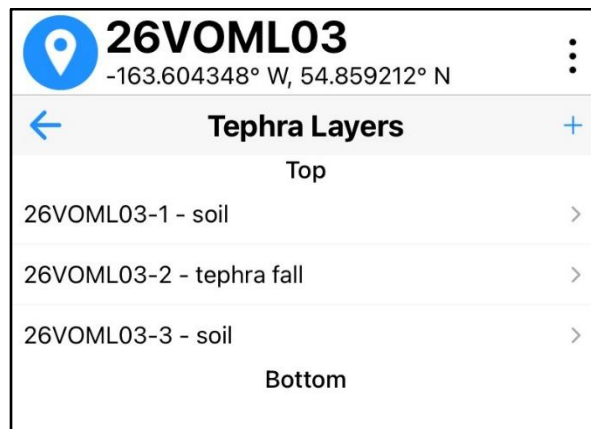
- Save!
9. Use the spot tool to annotate the next layer. If naming conventions are setup correctly, the spot name will be station-02, -03, and so on. Layer label will always be this spot name with “-1” on the end, you can delete this now or after you extract your data from the app.
 10. You can add a sample to any layer using the “Sample” tool (rock hammer icon). You could also just add sample info to the layer description (“sampled bulk tephra as “A”).

Tephra Layers Option 2 (log layers in order)

1. If you do not have an image of your section, or want to quickly log without annotating an image (images can still be saved, but you are not using them as a “basemap”), you can also just log a whole section within a station by adding multiple layers to the station with the volcano icon.
 - At AVO we often use numbered pins to annotate the outcrop (and thus photos of it) and pre-number the layers. This format allows simply describing each layer associated with its number.



2. Add the first layer with the “+” icon as described in #6 above.
3. After you save the layer, use the “+” icon to add another layer and layer metadata.
4. By default, layers are added from top to bottom, but you can manually reorder them by drag and drop. The layers keep their labels, so the numbering in the “layer label” stays as originally entered.



Data Export

1. Backup to your device: When you are done for the day, or even with a section, it's always a good idea to backup the project to your device. Having a local saved snapshot of your project

is a lifesaver if the app has a glitch, and it's a free app, so you should expect bugs and glitches to happen!

- In the Home menu (left side), under “MANAGE PROJECT” choose “Backup”
 - “Save” will save locally on your mobile device in the “ProjectBackups” folder
 - “Save and zip” will also save locally, but as a .zip in the “Distribution” folder

2. Upload to the server:

- With an internet connection, use the backup option to sync your project with the online server.
- In this same menu (Home > Backup), “Upload” will upload the project to the server, overwriting older datasets with any newer ones on your device.
- If you ONLY edited your own dataset (all others “read-only” or untouched) you shouldn't have a problem syncing, as long as all project members only edited their own dataset.
- If you added data to the same dataset as someone else, the last edits will overwrite earlier ones and errors are very likely. ****Only make edits to your own dataset**** (best practice: set all other datasets to “read-only”)

3. View and download your data from the StraboSpot website:

- Once synced online, login to the StraboSpot server and go to “My StraboSpot Field Data”.
- In your project you should see spots under the dataset you added to.
- Choose “download” and the options include an XLS option.
 - You can extract all the layer data from the XLS
 - You can also download PDFs of the “field notebook” shapefiles of the stations, and the annotated imagery.
 - We are hoping to have a tephra specific download option in the near future.